

CIA1 MLOps EXP Assignment (20 Marks)

Assignment Title:

Design and Implement a Basic MLOps Pipeline for an ML Application using anyone unique dataset of your choice.(Specify data set name and ML models in shared sheet)

Objective

To understand the practical implementation of MLOps concepts including experiment tracking, data versioning, workflow management, feature engineering, and cloud deployment considerations.

Problem Statement

You are provided with a machine learning dataset (Eg: Any data set for ML Classification with Artificats/Large Data Sets). Design a simple MLOps workflow that manages the complete lifecycle of the model from data preparation to deployment readiness.

Perform the following tasks:

Task 1: MLOps Lifecycle Analysis (4 Marks)

1. Draw the end-to-end ML lifecycle for the selected dataset.
2. Identify:
 - Data ingestion stage
 - Data preprocessing stage
 - Model training stage
 - Model evaluation stage
 - Deployment stage
 - Monitoring stage

Task 2: Experiment Tracking using MLflow (4 Marks)

1. Train at least two machine learning models(like KNN, SVM,RF etc.,)
2. Track:
 - Parameters
 - Training accuracy

- Testing accuracy
 - Precision/Recall/F1-score
3. Compare the experimental results using MLflow.

Task 3: Dataset Versioning with DVC (4 Marks)

1. Initialize a DVC repository.
2. Add the dataset to DVC.
3. Create two versions of the dataset:
 - Original dataset
 - Modified dataset with added preprocessing
4. Compare both versions and explain the benefits of data versioning.

Task 4: Workflow Automation Design (4 Marks)

1. Design a workflow pipeline using Kubeflow concepts.
2. Create a pipeline consisting of:
 - Data Collection
 - Data Validation
 - Training
 - Evaluation
 - Deployment
3. Give your explanation on how automation improves reproducibility and scalability.

Task 5: Feature Store and Cloud Platform Study (4 Marks)

Part A: Feast Feature Store

1. Identify three features used in your ML model.
2. Explain how Feast can manage and serve these features consistently during training and inference.

Part B: Cloud Platform Comparison

Compare any two of the following platforms for MLOps implementation/or launch your MLOPs in GCP

- AWS SageMaker
- Google Cloud Vertex AI

- IBM Watson Studio

Based on:

- Services offered
- Experiment tracking
- AutoML support
- Deployment options
- Cost considerations

Marking Scheme (20 Marks)

Component	Marks
ML Lifecycle Design	4
MLflow Experiment Tracking	4
DVC Version Control	4
Kubeflow Workflow Design	4
Feast & Cloud Platform Analysis	4
Total	20

Note: Implement the complete pipeline using:

- MLflow for experimentation
- DVC for dataset versioning
- Kubeflow Pipeline (or simulated workflow)
- Deployment on AWS SageMaker / Google Vertex AI

Prepare a **5-minute demo presentation** showing the end-to-end MLOps workflow.

Learning Outcomes:

After completing this assignment, students will be able to:

- ✓ Explain the evolution and significance of MLOps

- ✓ Differentiate DevOps and MLOps practices
 - ✓ Address ML production challenges
 - ✓ Manage the ML lifecycle effectively
 - ✓ Use MLflow and DVC for reproducibility
 - ✓ Design scalable workflows using Kubeflow
 - ✓ Understand feature stores using Feast
 - ✓ Evaluate cloud platforms for enterprise MLOps solutions.
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